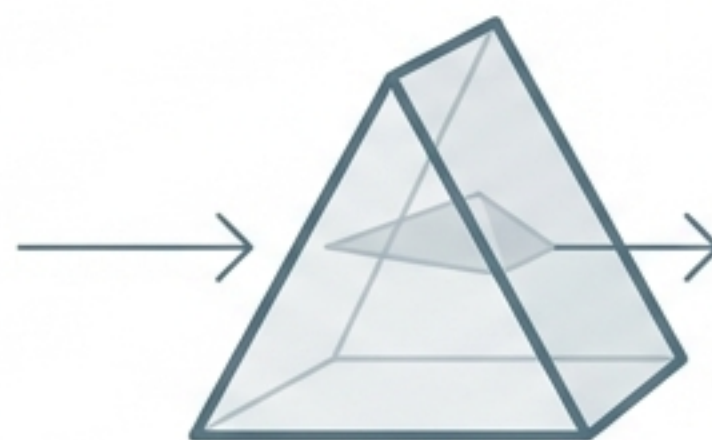


From Strings to Meaning: The Evolution of Vector Semantics

How computational linguistics moved from counting symbols to predicting context.

banking (n.)
The business conducted or
services offered by a bank.



$[-0.34, -0.84, 0.20, -0.26, -0.12,$
 $0.23, 1.04, -0.16, 0.31, 0.06 \dots]$

1. Lexical Semantics:
Discrete Symbols

2. Sparse Vectors:
The Era of Counting
(TF-IDF)

3. Dense Embeddings:
The Era of Prediction
(Word2Vec, GloVe)

4. Visualization:
Mapping Meaning

The Problem of Discrete Representation

Lexical Semantics

Traditionally, NLP treated words as atomic symbols. In a localist representation, “hotel” and “motel” are distinct IDs with no inherent mathematical relationship.

hotel

motel

One-Hot Vectors & Orthogonality

motel = [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0]

hotel = [0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0]

$$\text{motel} \cdot \text{hotel} = 0$$

Orthogonal vectors share no similarity. To the computer, “motel” is as different from “hotel” as it is from “hamburger”.

The Distributional Hypothesis

“You shall know a word by the company it keeps.” — J.R. Firth (1957)

He filled the **[wampimuk]** with alcohol.



We sat around the camp fire and drank **[wampimuk]**.



Inference: **wampimuk** is a vessel for liquid AND an alcoholic beverage.

Meaning is not contained within the symbol, but defined by its environment.

The Era of Counting: Term Frequency (TF)

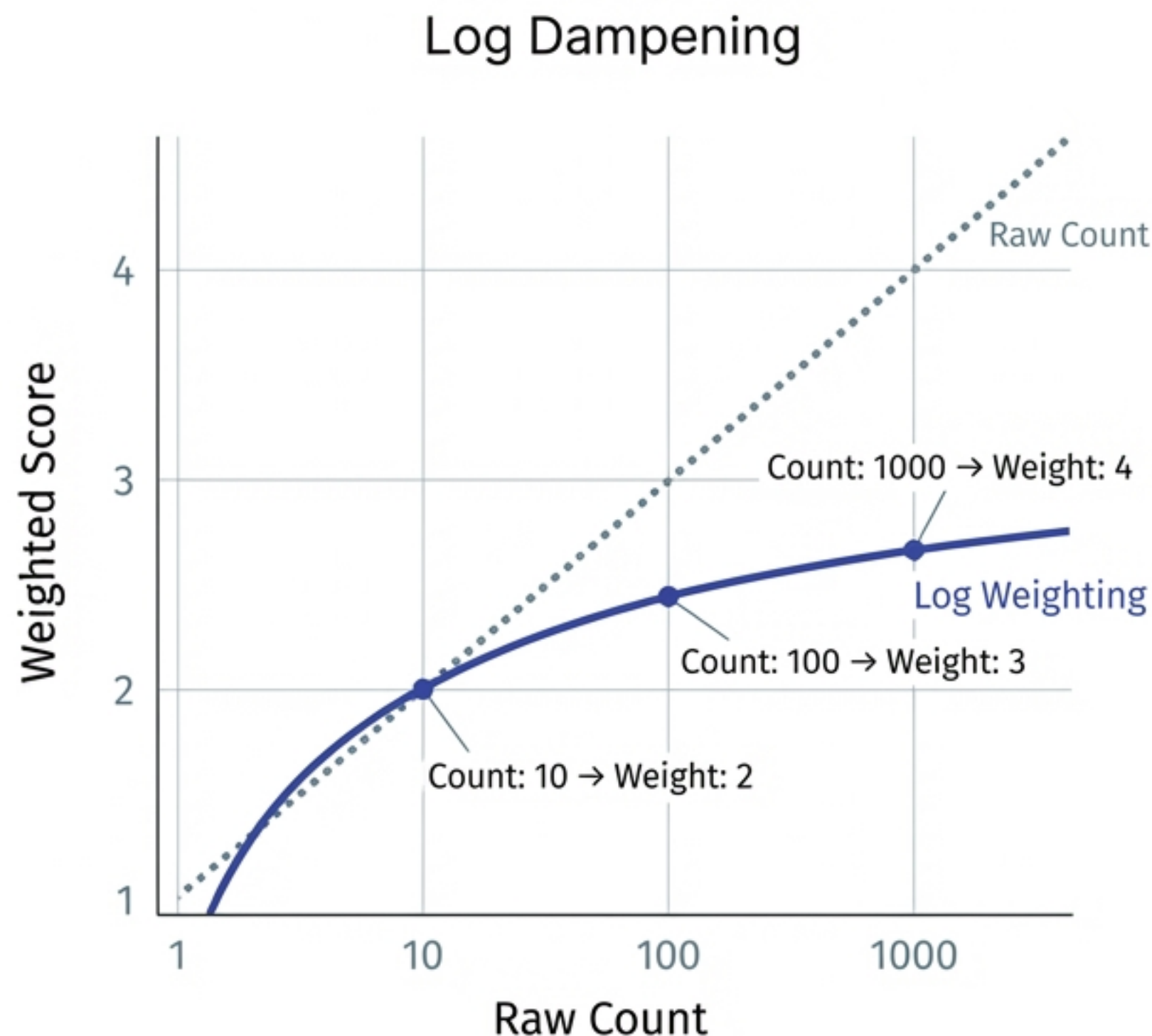
Term Frequency (TF):

A raw count of how many times word t appears in document d .

$$tf(t,d) = 1 + \log_{10}(\text{count}(t,d))$$

The Problem:

Relevance implies **diminishing returns**. A word appearing 100 times is **not 100x** more relevant than a word appearing once.



Balancing Frequency with Distinctiveness (TF-IDF)

Inverse Document Frequency (IDF):

We must penalize words that appear in every document (like 'the', 'good') and boost rare, distinctive words.

$$\text{idf}(t) = \log_{10}(N / \text{df}(t))$$

where N = Total Documents, $\text{df}(t)$ = Documents containing term t .

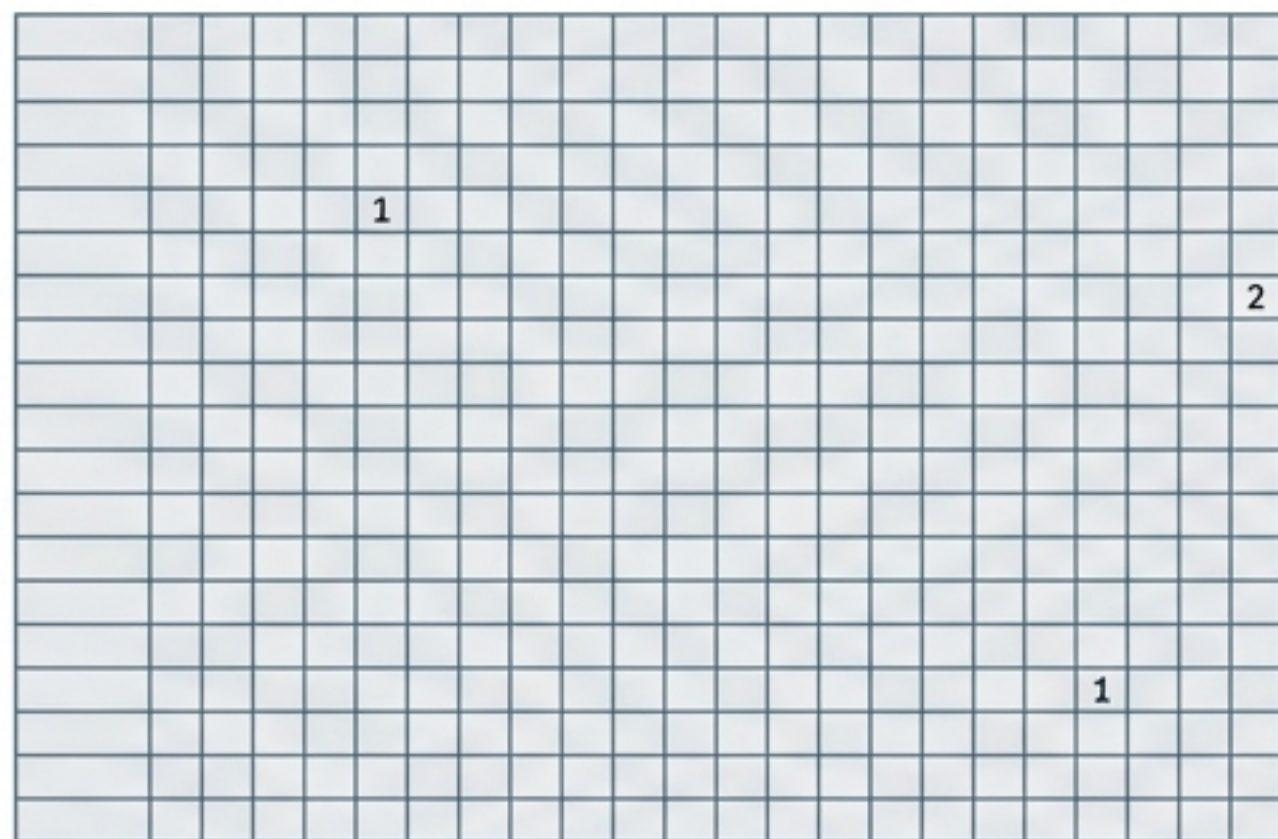
The Shakespeare Nano-Example ($N=37$ Plays)

Word	Df (Plays containing word)	IDF Score	Interpretation
Romeo	1	1.57	Highly Distinctive
Salad	2	1.27	Moderate
Forest	12	0.489	Common
Sweet	37	0.0	Ignored (Stop word)

Final Weight = $\text{tf} \times \text{idf}$. High weight means the word is frequent in *this* document, but rare in *others*.

The Shift from Sparse to Dense Vectors

Sparse Matrix (TF-IDF)



- Dimensions: $|V|$ (e.g., 50,000)
- Values: Discrete Counts
- Status: Mostly Zeros (**Inefficient**)

The Insight:
Don't count neighbors.
Train a classifier to
PREDICT neighbors.



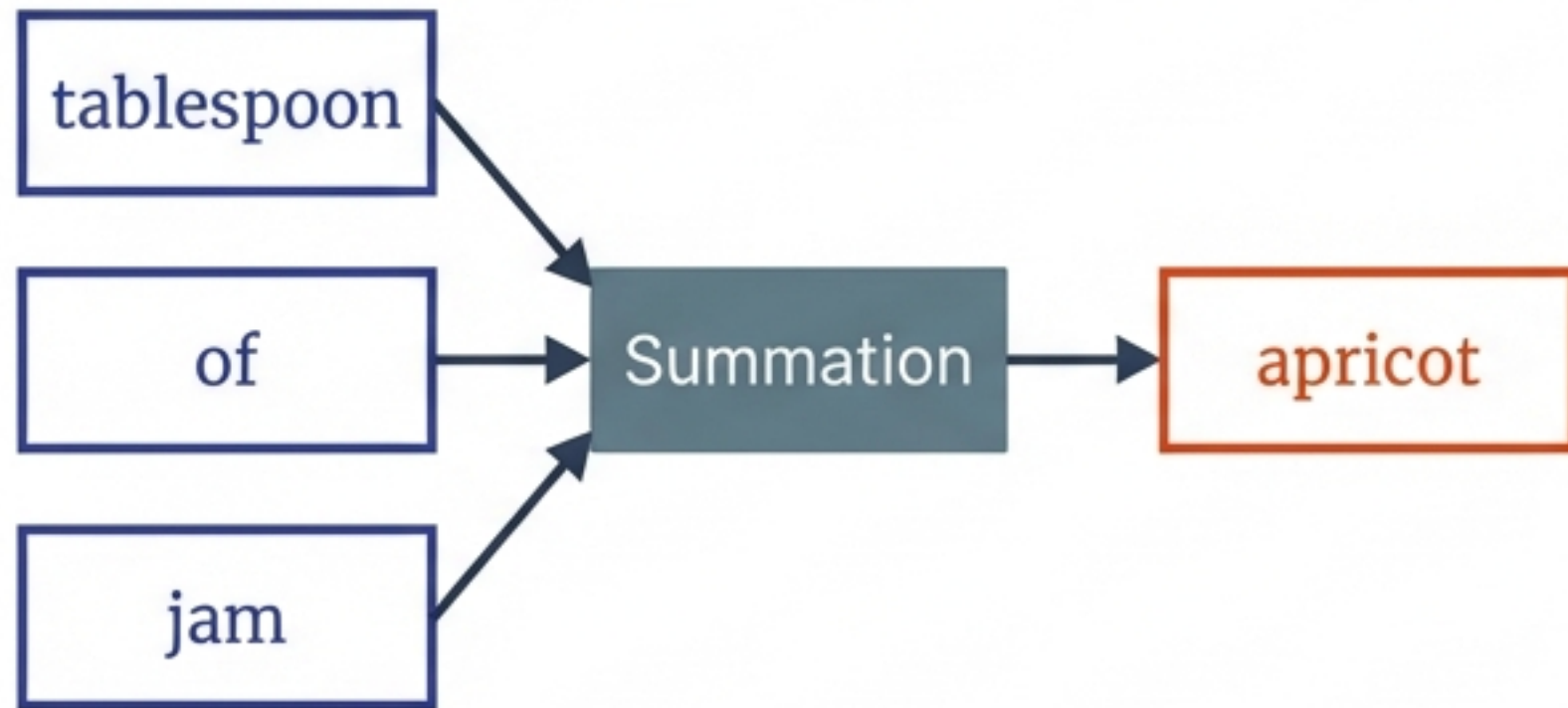
Dense Vector (Embeddings)



- Dimensions: d (e.g., 300)
- Values: Continuous Reals (Weights)
- Status: Dense Information

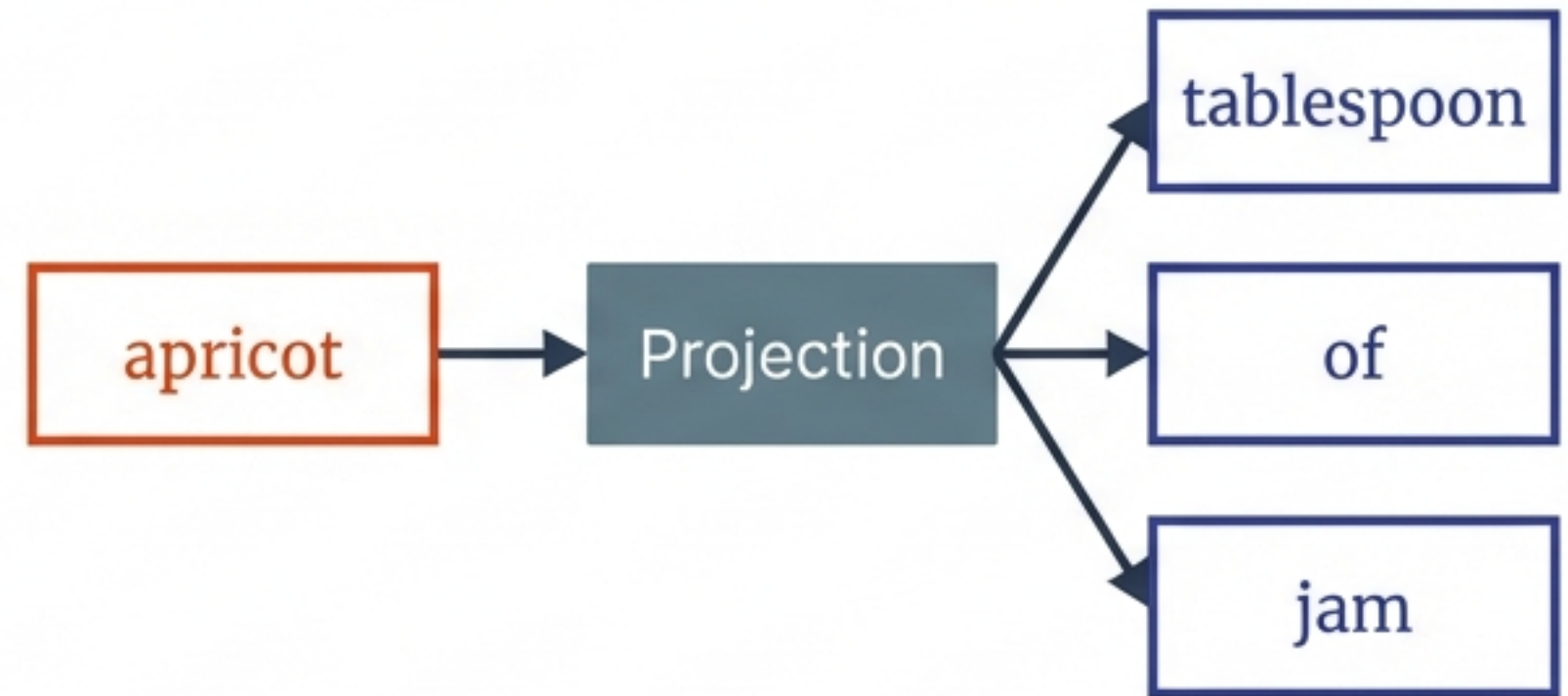
Word2Vec Architectures: CBOW vs. Skip-gram

CBOW (Continuous Bag of Words)



Predict Target from Context.

Skip-gram



Predict Context from Target.

We will focus on Skip-gram, which works better for semantic details and larger corpora.

Deep Dive: The Skip-gram Mechanism

The Sliding Window

...lemon, a tablespoon of [apricot] jam, a pinch...

Training Pairs

(apricot, tablespoon)
(apricot, of)
(apricot, jam)
(apricot, pinch)

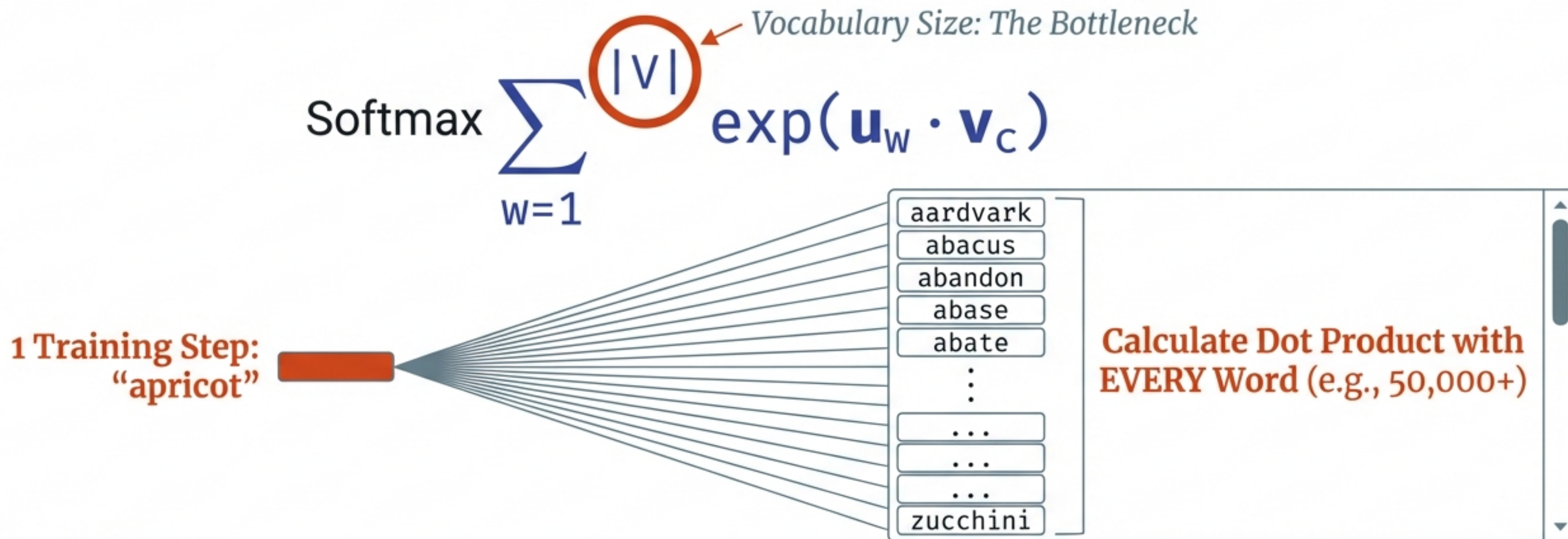
The Probability (Softmax)

$$P(o|c) = \frac{\exp(u_o \cdot v_c)}{\sum (\exp(u_w \cdot v_c))}$$

Dot product (Similarity) between
Target and **Context** vectors.

Sum of similarity to **ALL**
words in vocabulary.

The Complete Computational Bottleneck



The Problem:

To update weights for just **ONE word** (e.g., "apricot"), we must calculate the dot product with **EVERY word** in the dictionary (aardvark, abacus, ... zucchini).

Implication:

If **Vocabulary** = 100,000, this is prohibitively slow.

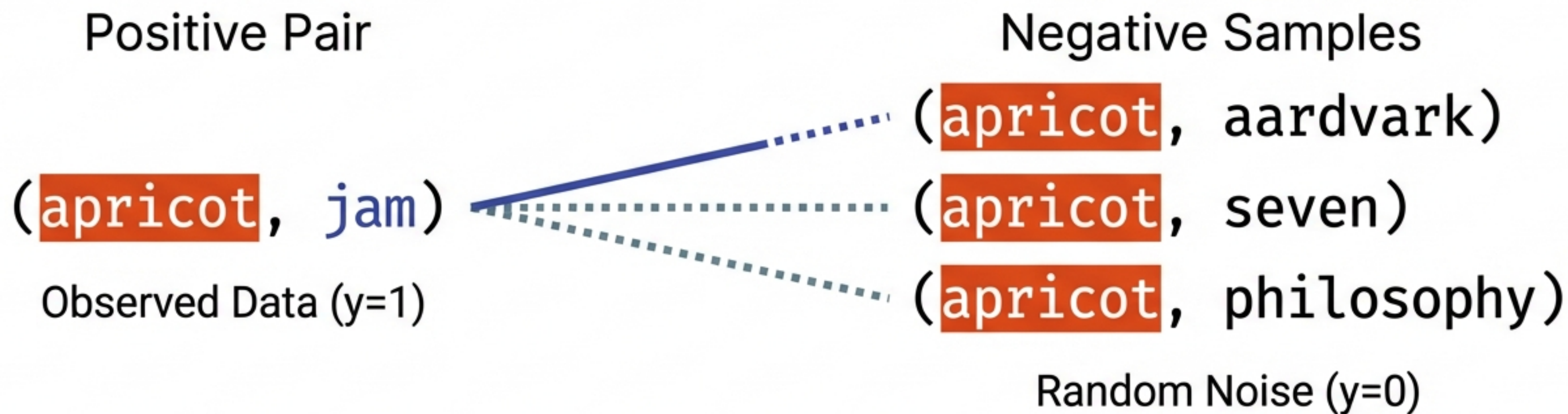
Solution Needed:

Approximation.

Efficiency: Skip-Gram with Negative Sampling (SGNS)

The Insight: Reformulate as a Binary Classification problem.

Instead of predicting a full distribution, simply ask: Is this pair real or noise?



$$\text{Maximize: } \log \sigma(\mathbf{u}_{\text{jam}} \cdot \mathbf{v}_{\text{apricot}}) + \sum \log \sigma(-\mathbf{u}_{\text{noise}} \cdot \mathbf{v}_{\text{apricot}})$$

We only update the target and k noise words. No summation over the whole vocabulary!

GloVe: Global Vectors for Word Representation

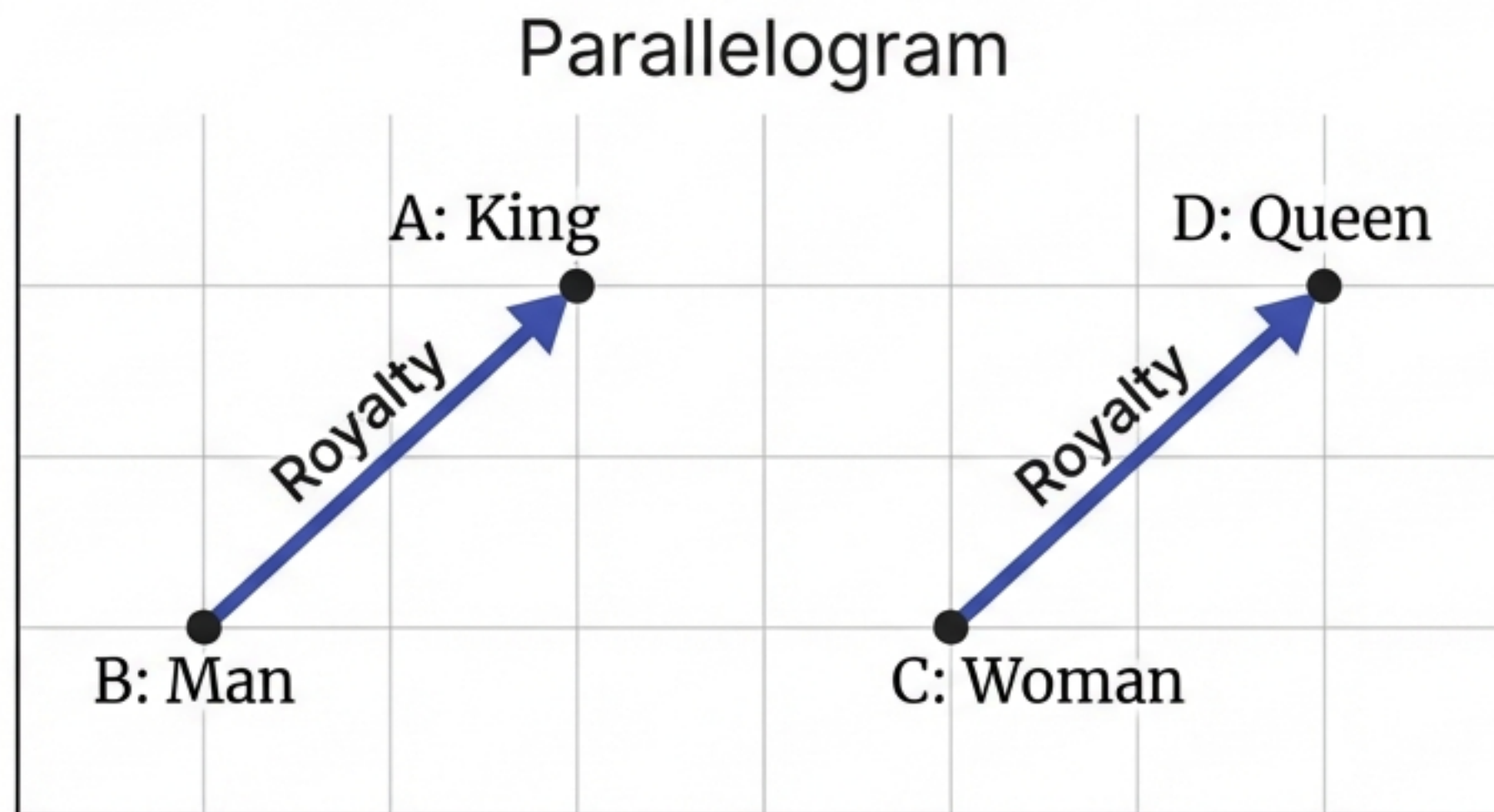
Combining the global statistics of Matrix Factorization with the local context of Word2Vec.

Ratio of Co-occurrence Probabilities

Probe Word (k)	$P(k \text{ice})$	$P(k \text{steam})$	Ratio $P(k \text{ice})/P(k \text{steam})$	Meaning
solid	High	Low	Large	Related to Ice
gas	Low	High	Small	Related to Steam
water	High	High	~ 1	Related to Both
fashion	Low	Low	~ 1	Unrelated

Key Insight: GloVe explicitly models these ratios. It learns vectors such that their **dot product** equals the **log** of their **probability of co-occurrence**.

Analogical Reasoning and Vector Arithmetic



$$\mathbf{Vector}(\text{King}) - \mathbf{Vector}(\text{Man}) + \mathbf{Vector}(\text{Woman}) \approx \mathbf{Vector}(\text{Queen})$$

Paris – France + Germany $\approx$ Berlin (Capital–Country) in Merriweather
Walked – Walk + Fly $\approx$ Flew (Verb Tense) in Merriweather Regular

Visualizing Embeddings in Inter Bold

Dimensionality reduction (t-SNE/PCA) allows us to project 300-dimensional vectors onto a 2D plane to see semantic clusters.



The algorithm grouped these concepts automatically based on distribution, without knowing their definitions.

Summary & Frontiers

Inter Bold, Charcoal (#1A1A1A)

Discrete Era

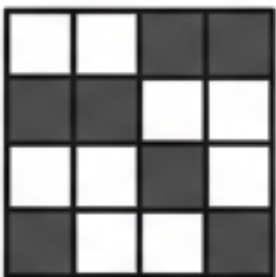
Inter Medium



One-Hot Vectors.
No meaning.
Merriweather
Regular

Sparse Era

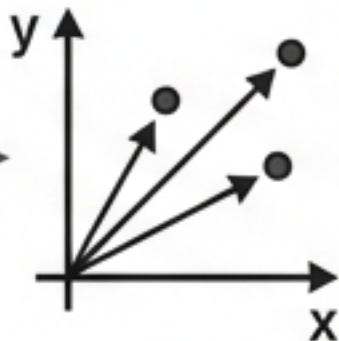
Inter Medium



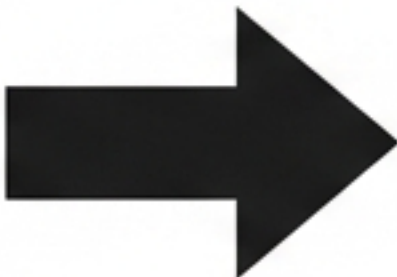
TF-IDF.
Meaning via
Counting.
Merriweather
Regular

Dense Era

Inter Medium



Word2Vec/GloVe.
Meaning via
Prediction. Static
Embeddings.
Merriweather Regular



The Next Frontier: Contextual Embeddings

Inter Medium, Charcoal

Static embeddings assign one vector to “stick” regardless of context. Modern models (ELMo, BERT) dynamically change the vector based on the sentence.

